CODE: -

```c
#include <stdio.h>

#include <math.h>

// Recursive function for T(n) = T(n/2) + log2(n)

double T(int n)

{

    if (n <= 1)

        return 1;

    return T(n / 2) + log2(n);

}

int main()

{

    int n;

    printf("Enter value of n: ");

    scanf("%d", &n);

    double result = T(n);

    printf("\nT(%d) = %.2lf\n", n, result);

    printf("Asymptotic Complexity = Theta((log n)^2)\n");

    return 0;
```

Output

```
Enter value of n: 16

T(16) = 11.00
Asymptotic Complexity = Theta((log n)^2)


=== Code Execution Successful ===
```